Decoding the Card Trick:

Autonomous Strategy Discovery via Reinforcement Learning

by

Meenakshisundram Ganapathi Subramanian

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Zhe Xu, Chair
Spring Berman
Hamidreza Marvi

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ABSTRACT

Robots functioning in unstructured environments often need to frequently interface with dynamic systems where full state observability is limited or intentionally unobservable. This thesis seeks to explore the topic of autonomous strategy discovery and state inference from the standpoint of a well-known human-robot interaction (HRI) benchmark: The Three-Card Deception. Implemented via a Stretch 3 mobile manipulator robot, the system must dynamically infer a human participant's secretly chosen card from a set of deterministic spatial permutations.

To address this issue, a novel two-stage Reinforcement Learning (RL) pipeline is proposed in this thesis, which completely eliminates the requirement for a predefined heuristic strategy. In the first place, a Strategy Discovery Agent (SDA) with a Proximal Policy Optimization (PPO) approach will be utilized to explore the underlying physical mechanisms of the trick, which enables the agent to learn the optimal sensing action autonomously without ambiguity, i.e., the exact spatial index can be controlled without ambiguity. In the second place, a State Inference Agent (SIA) with computer vision, which uses the object detection approach of You Only Look Once (YOLO), will be used to logically infer the hidden initial choice of the human.

Through the modeling of the physical environment as a Partially Observable Markov Decision Process (POMDP), the system can guarantee the discovery of a strategy that provides perfect information. This framework demonstrates the effectiveness of sequential use of PPO networks in connecting physical manipulation with logical deduction, thereby showing a robust architecture for autonomous decision-making in a physical environment. Moreover, the use of a Large Language Model (LLM) enables a responsive and autonomous 'magician' persona, making the benchmark from a physical manipulation scenario to a human-robot interaction scenario.

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Chapter 1

INTRODUCTION

As robots increasingly move from highly structured, controlled environments in industry into dynamic, unstructured, and highly uncertain human environments[1], the need for reliable decision-making becomes critical. Most current robots use highly structured, highly observable state models as a means of performing functions. However, in a highly uncertain environment, there will always be unknowns, hidden variables, and states that are either intentionally or inadvertently obscured. In order for a robot to become a highly autonomous entity, not only will it need to be able to act on the information that is available, but it will also need to be able to autonomously devise a method for attaining the information that it lacks.

This thesis approaches the challenge of autonomous information gathering and state inference through a classic human-robot interaction benchmark: the Three-Card Deception. While traditionally rooted in parlor tricks and the study of robot deception [2], this scenario serves as a rigorous mathematical and physical benchmark for robotics, encompassing perception, manipulation, logical reasoning, and Human-Robot Interaction (HRI). Solving the puzzle requires the robot to bridge the gap between physical action in the real world and abstract logical deduction in its software architecture.

In order to tackle this challenge, this thesis introduces a new and unprecedented method for strategy discovery in physical logic puzzles without relying on heuristics. This method allows for learning to act optimally through environmental interaction. The system follows a decoupled two-stage Reinforcement Learning (RL) paradigm. The first stage involves a strategy discovery agent. This agent is used to explore the trick and learn to act optimally by utilizing Proximal Policy Optimization (PPO) [3]. The second stage involves a state inference agent. This agent infers the state of the user based on visual observations.

This decision-making framework is backed by a strong integration for perception and execution, where a YOLO-based computer vision pipeline [4] is fine-tuned for a reliable state estimation during physical manipulation. Finally, the end-to-end pipeline is implemented on a Stretch 3 mobile manipulator [1] with a Large Language Model (LLM) [5] for creating a dynamic interactive robotic persona that guides the human participant through the deception scenario, advancing the system beyond a manipulation scenario to a more engaging autonomous HRI framework.

The rest of this thesis is organized as follows: Chapter 2 describes the related work in the areas of robotic manipulation, reinforcement learning for strategy discovery, and human-robot interaction. Chapter 3 describes the problem statement where the physical trick is formulated as a Partially Observable Markov Decision Process. Chapter 4 describes the methodology where the agents for Strategy Discovery and State Inference are trained. Chapter 5 describes the experiments where the ROS 2 system architecture, hardware execution, and evaluation are discussed. Chapter 6 concludes the thesis with a discussion on future work.

Chapter 2

RELATED WORK

This chapter discusses the existing literature relevant to this thesis. The discussion is divided into four major areas of focus. These areas include the use of reinforcement learning in robot decision-making, active perception and partial observability, computer vision for robust state estimation, and the use of Large Language Models (LLMs) in Human-Robot Interaction (HRI).

2.1 Reinforcement Learning in Robotic Manipulation

Deep Reinforcement Learning (DRL) has become a promising approach to address complex robot manipulation tasks traditionally requiring extensive mathematical modeling and programming heuristics [6, 7, 8]. Algorithms such as Proximal Policy Optimization (PPO) [3] have become popular due to their reliable convergence and sample efficiency. Although most of the DRL research was initially dedicated to low-level motor control and grasping tasks[9], current trends favor higher-level task planning and strategy learning. Unlike traditional approaches where the robot is hardcoded to perform a series of pre-programmed states, researchers are now using Partially Observable Markov Decision Processes (POMDPs) to represent logic puzzles and sequential manipulation tasks. This allows the robot to autonomously learn the optimal logic required to solve a manipulation task through environmental interaction [10]. In this thesis, a decoupled two-stage PPO algorithm is used to autonomously learn the physical logic required to solve a deception game.

2.2 Active Perception and Partial Observability

In unstructured real-world environments, it is rarely the case that a robot has full knowledge of its surroundings. This problem is typically modeled under a Partially Observable Markov Decision Process (POMDP) [11]. To function correctly in an unstructured real-world environment, a robot must engage in *active perception*—the process of taking physical actions to gather information about the environment [12]. Traditionally, active perception methods have been based on human-engineered heuristics and computationally expensive belief state planning. However, recent advances have aimed to learn active perception strategies end-to-end [13]. The Three-Card Deception is an adversarial benchmark for active perception. The robot must find a single physical action to reveal a hidden variable despite spatial permutations.

2.3 Vision-Based State Estimation

However, the existing gap between the real world and the discrete mathematical spaces necessary for the functioning of RL agents can only be filled by the implementation of robust computer vision pipelines. Object detection pipelines, specifically the YOLO (You Only Look Once) series of computer vision models [4], have been established as the de facto standard for real-time object tracking for robotics applications owing to their high speeds and accuracy. Nevertheless, the raw video feed for a dynamic manipulation task is often plagued by motion blur, occlusion, and flickering effects, which can have a devastating impact on the state inference capabilities of the RL agent[14]. In this regard, the use of temporal filtering and multi-frame consistency-based computer vision pipelines is often necessary to improve the robustness of the video feed before it is forwarded to the decision-making component. This work avoids the use of filtering-based computer vision pipelines and instead relies on a fine-tuned YOLO pipeline to address the domain shift. A pre-trained card detection computer vision model was fine-tuned using a hybrid dataset comprising both standard playing card images and custom images obtained from the physical workspace of the Stretch 3 robot.

2.4 LLMs in Human-Robot Interaction

The integration of Large Language Models (LLMs) in robotic systems is a game-changer for Human Robot Interaction (HRI) [15] systems. The traditional dialogue systems for robots are based on rigid decision trees, making it difficult for users to have a natural conversation. The inclusion of foundational models in robots allows for natural language processing, understanding, and generating responses [5, 16, 17]. In deception games, it is as important how you deliver a message or act as it is what you are doing. The inclusion of audio processing and text-to-speech systems enables modern robots to have an engaging persona. This thesis aims to utilize this feature of modern robots to convert a mobile manipulator, Stretch 3, into an interactive magician.

Chapter 3

PROBLEM FORMULATION

In this chapter, we formalize the physical mechanics of the Three-Card Deception, and we express the autonomous strategy discovery problem as a mathematical framework. We begin by expressing the physical mechanics of the problem as a Partially Observable Markov Decision Process (POMDP) [11] and subsequently express it as a decoupled reinforcement learning problem. This formalization provides the basis for the heuristic-free framework discussed in Chapter 4.

3.1 Mechanics of the Three-Card Deception

The physical environment is composed of three cards, represented as spatial indices $I = \{0, 1, 2\}$, each of which is initialized with a value $v_i \in \{0, 1, 2\}$ such that $v_i = i$ for all $i \in I$. The physical mechanics of the problem, from the initial state through the user's choice and into the unknown swap, are represented in Figure 3.1.

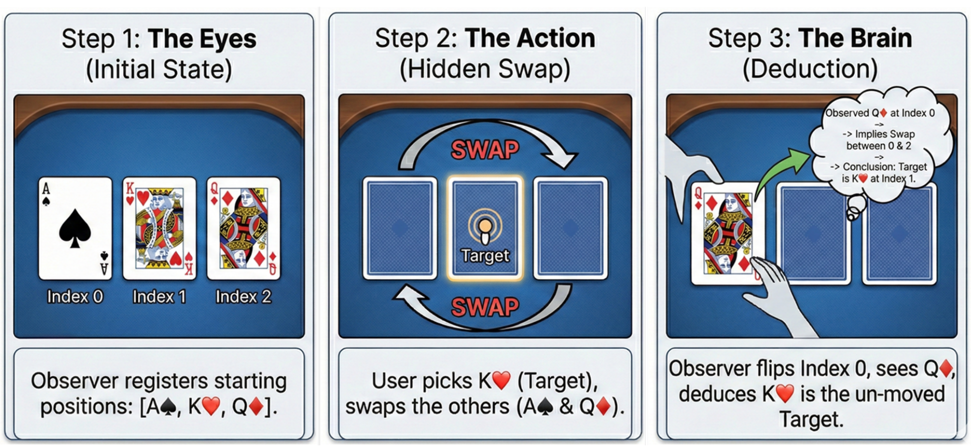

Figure 3.1: Mechanics of the Three-Card Deception: The physical sequence of the environment, demonstrating the initial state, the hidden swap of the unchosen cards, and the final deduction achieved by flipping a single card to resolve ambiguity.

The problem begins with an initialization step, during which a human user selects a card of interest. We represent this unknown human choice as $u \in I$, and we assume the robotic system is unaware of u.

After the user's choice, a deterministic spatial permutation that conceals information about the hidden state occurs in the environment. This occurs as a result of a swap in the positional value of the two cards that were not selected by the user. Let the remaining two indices be j and k, where $j, k \in I \setminus \{u\}$ and $j \neq k$. A deterministic swap of the resident values occurs as follows:

$$v_j \leftarrow v_k \quad \text{and} \quad v_k \leftarrow v_j \tag{3.1}$$

However, the challenge for the robotic system lies in inferring the hidden variable u through a physical sensing action, where a card at a certain index is flipped in order to reveal its value.

3.2 The Perception-Planning Problem

Since the initial state is unknown, from the robot's perspective, the environment is partially observable. The goal is to find a static optimal sensing action $a^* \in I$ that mathematically ensures the inference of u, irrespective of the choice of card chosen by the human. The system, through a physical action taken to reduce uncertainty about the environment, is equivalent to performing an autonomous active perception strategy[12, 18].

In order for a heuristic-free solution to be achieved, logical rules relating visual information to the unknown choice of the human are required. Although the simulator of the agent models the physical mechanics of the trick (spatial permutations), it is required for the system to discover a^* autonomously based on the evaluation of pure information gain.

3.3 Partially Observable Markov Decision Process (POMDP) Formulation

To solve the perception planning problem via reinforcement learning, the strategy discovery is modeled as a Partially Observable Markov Decision Process (POMDP). Unlike a standard MDP where the environment is fully visible, a POMDP accounts for hidden variables—in this case, the face-down cards. The POMDP is defined by the tuple $(\mathcal{S}, \mathcal{A}, \mathcal{T}, \mathcal{R}, \Omega, \mathcal{O})$, where $\mathcal{S}$ represents the hidden states, $\mathcal{A}$ is the action space, $\mathcal{T}$ represents deterministic state transitions, $\mathcal{R}$ is the reward function, Ω represents the observation space (the visual data retrieved by the camera), and $\mathcal{O}$ represents the observation probabilities. This POMDP is particularly designed for the Strategy Discovery Agent to evaluate the utility of different physical manipulations in revealing the hidden state.

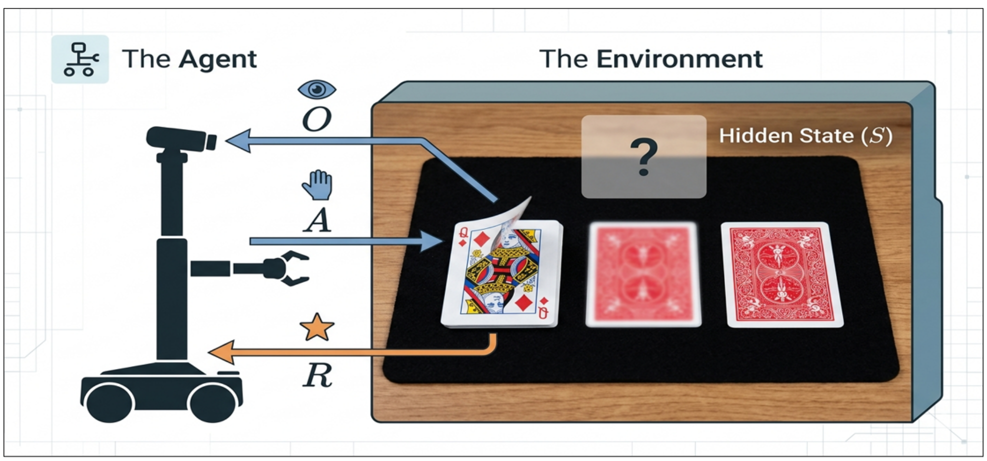

Figure 3.2: POMDP Architecture: The interaction cycle between the robotic agent and the partially observable environment, illustrating the flow of physical actions ($\mathcal{A}$), resulting visual observations (Ω), and evaluated rewards ($\mathcal{R}$) over the hidden state ($\mathcal{S}$).

3.3.1 State Space

Given that the physical mechanics of the Three-Card Deception are deterministic and that the optimal sensing strategy is static over all possible episodes, the environment does not need complex state tracking for the discovery phase. As a result, the state space $\mathcal{S}$ is defined as a continuous, one-dimensional dummy variable:

$$\mathcal{S} \in \mathbb{R}^1 \tag{3.2}$$

where the state is consistently initialized as $s = [0.0]$.

3.3.2 Action Space

The action space $\mathcal{A}$ is defined by the physical decision regarding which card the robot will manipulate (flip) for a visual observation. This is defined as a discrete space based on the spatial indices of the cards:

$$\mathcal{A} \in \{0, 1, 2\} \tag{3.3}$$

3.3.3 Observation Space

In a POMDP, the agent does not have direct access to the hidden state $\mathcal{S}$. Instead, it receives an observation $o \in \Omega$ that provides partial information about the environment following an action.

For the Three-Card Deception framework, the observation space Ω is defined by the visual data registered by the YOLO-based computer vision pipeline after a physical flip action is executed. Specifically, when the robot manipulates a target index, the camera pipeline processes the exposed card and returns a discrete integer value $o \in \{0, 1, 2\}$ corresponding to the card's identity. This visual observation acts as the critical bridge between the physical workspace and the logical deduction framework, providing the raw data necessary to evaluate the information gain and ultimately resolve the hidden user choice.

3.3.4 Reward Function

The reward function $\mathcal{R}(s, a)$ is the core component of the entire process of heuristic-free discovery. The reward function is intended to strongly penalize ambiguity and reward a strategy perfectly.

For a given sensing action a, the agent "simulates a multiverse outcome over all possible user choices $u \in \{0, 1, 2\}$." The observation o_u is defined as the observation of the card at index a given that the user has chosen u. The agent receives a set of observations O:

$$O = \{o_0, o_1, o_2\} \tag{3.4}$$

The reward is granted based strictly on the cardinality of unique observations in the set O:

$$\mathcal{R}(s, a) = \begin{cases} +10.0 & \text{if } |O| = 3 \\ -10.0 & \text{otherwise} \end{cases} \tag{3.5}$$

A maximum reward is only attainable if the observation O resulting from the chosen action a has a cardinality of 3, meaning $|O| = 3$, and thus a bijection exists between O and the possible user choices.

3.4 State Inference Formulation

Once the Strategy Discovery Agent has converged to the optimal physical action a^*, the secondary State Inference Agent has to learn the inverse of the mapping function. The State Inference Agent has to convert the observed visual information back into the hidden user state.

In the formulation of the State Inference problem, the observed value of the card $o \in \{0, 1, 2\}$ is the continuous input state, and the agent has to make a discrete prediction $\hat{u} \in \{0, 1, 2\}$ that corresponds to the user's initial choice. The reward for the State Inference problem is sparse: $+1.0$ if the prediction is correct, $\hat{u} = u$, and -1.0 if the prediction is incorrect.

Chapter 4

METHODOLOGY

The Partially Observable Markov Decision Process (POMDP) is developed in the previous chapter. This chapter outlines the implementation details of the decoupled reinforcement learning pipeline. The physical mechanics of the Three-Card Deception are successfully replicated through simulation, and two separate agents are developed to learn the physical strategy and logical states independently.

4.1 Reinforcement Learning Architecture

The core decision-making framework is developed using the Stable Baseline3 library [19], employing the PPO algorithm. The PPO algorithm was chosen for its reliable performance, convergence, and sample efficiency in both discrete and continuous action space.

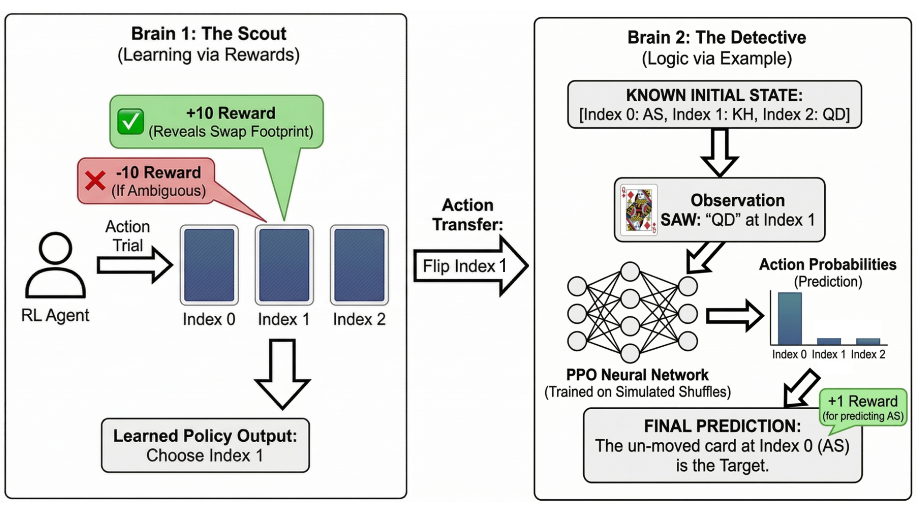

Figure 4.1: Two-Stage RL Architecture: The decoupled pipeline where Brain 1 (The Scout) discovers the optimal physical manipulation via a multiverse reward structure, and Brain 2 (The Detective) learns the logical deduction mapping.

Instead of attempting to train a single neural network to perform both physical and logical tasks simultaneously, the proposed solution is decoupled into a two-stage pipeline. This decouples the complexity of the state and action space, allowing for faster training and a clean transition between the physical and logical agents. The overview of the two-stage training and inference pipeline is shown in Figure 4.1.

4.2 Stage 1: Strategy Discovery Training

The first step of this pipeline is to train the Strategy Discovery Agent for identifying the best placement of sensors, i.e., which card index needs to be physically manipulated in order to eliminate all ambiguity regarding the hidden user state.

4.2.1 Environment Implementation

The training environment was implemented as a custom OpenAI Gymnasium environment [20]. The environment encapsulates the `simulate_3card_physics` function, which mimics the deterministic swap of the two cards not chosen. To ensure heuristic-free learning, the agent is never provided with the underlying logic of this swap mechanism. Instead, it is only provided with the resultant state.

4.2.2 The Multiverse Reward Evaluation

In this environment, the agent's policy network is trained on the evaluation of the discrete action space $\mathcal{A} \in \{0, 1, 2\}$, i.e., it is evaluating whether it should flip the card at index 0, 1, or 2. To determine the reward for a chosen action, a feasibility check is performed using a "multiverse" simulation.

In other words, for the selected index, the environment will iterate through all three possible initial human choices. The physical permutation is simulated for each of these choices, and the value of the card at the selected index is stored. The agent is provided a reward of +10.0 if and only if the cardinality of the set of observed values is equal to three. If the cardinality is less than three, it means that there are many human choices that all lead to exactly the same visual observation, i.e., ambiguity in the hidden state. In this case, the agent is provided a strict penalty of -10.0.

The Strategy Discovery Agent was trained using a Multi-Layer Perceptron (MLP) policy for 5,000 timesteps. The PPO agent was able to converge on the optimal single-flip strategy through purely exploratory environmental interaction, autonomously learning which spatial index perfectly disambiguates the hidden user choice.

4.3 Stage 2: State Inference Training

After the Strategy Discovery Agent converges on the optimal manipulation strategy, the second stage of the pipeline is responsible for training the State Inference Agent, which decodes the visual data.

The environment for the State Inference Agent is parameterized based on the optimal sensor index discovered in Stage 1. In this environment, defined under the Gymnasium library, the initial state is randomized for simulating a blind human choice. The environment runs the physical simulation and retrieves the single integer value at the optimal sensor index.

The single integer value is the continuous observation state for the PPO agent. The action space is a discrete choice, which is based on the prediction of the human's initial target. The reward is sparse, i.e., +1.0 for correct prediction and -1.0 for incorrect prediction. Since the agent from Stage 1 ensures a bijective mapping from the observation to the hidden state, the agent from Stage 2 converges rapidly on the inverse mapping function, learning to decode the visual observation into a perfect logical deduction.

4.4 Heuristic Baseline Implementation

To properly assess the benefits of the autonomous RL framework, a traditional heuristic-based approach was also developed and implemented as a reference. In the case of the standard Three-Card Deception problem, the deterministic nature of the swap logic enables a human programmer to easily and manually derive the solution. This was implemented as a static dictionary mapping the physical state directly to the inferred target state (e.g., mapping a perceived state at the `LEFT` index to a deduction of the `RIGHT` index).

Although this approach enables near-instantaneous inference through simple algorithmic lookups, it is completely lacking in any ability to adapt. This approach was the primary control method utilized within this thesis to illustrate the limitations of human-derived heuristics as the system is challenged with previously unseen N-card cyclic shifts.

4.4.1 Hardware and Computational Setup

The computational framework utilized to train the reinforcement learning agents and implement the software architecture was run natively upon a workstation. To ensure native compatibility with the Robot Operating System (ROS 2) and low-level hardware drivers for the Stretch 3 mobile manipulator robot, both the simulated training loops and the physical node execution were run natively within an Ubuntu Linux environment.

4.4.2 PPO Hyperparameters

The Proximal Policy Optimization (PPO) algorithm for both the Strategy Discovery Agent and the State Inference Agent was implemented using the stable baselines3 library [19]. The networks were implemented using standard MLPs for the policy networks. The discrete logic puzzles did not require complex vision feature extraction (CNNs), so the hyperparameters provided stable convergence.

Table 4.1 details the specific network and training configurations used for the simulated environments.

Table 4.1: PPO Training Hyperparameters. The specific configuration used for training the Stable Baselines3 agents across both the 3-card and 4-card environments.

Parameter	**Value**
Policy Architecture	`MlpPolicy`
Learning Rate (α)	3×10^{-4}
Batch Size	64
Number of Steps (`n_steps`)	2048
Discount Factor (γ)	0.99
GAE Lambda (λ)	0.95
Clipping Range	0.2
Number of Epochs	10

4.5 Model Deployment Strategy

After the successful execution of the training loop, the optimized weights and network architectures for both the Strategy Discovery Agent and the State Inference Agent are exported as static models. By freezing the network weights after training, the models can be used directly in the real-time robotic control architecture, where instantaneous inference is possible during human-robot interaction without the computational overhead of continuous learning.

Chapter 5

EXPERIMENTS

In order to verify the heuristic-free strategy discovery framework, the overall framework was executed on physical hardware. This chapter also explains the overall system architecture of the ROS 2 system, hardware setup using the Stretch 3 mobile manipulator, and the overall results. In addition, it also proves the scalability of the overall RL architecture by applying it to a more complex version of the deception, i.e., four cards.

5.1 ROS 2 System Architecture

The overall software stack connecting the trained RL models and physical robots is based on a Robot Operating System 2 (ROS 2) [21]. The overall system is highly modular, ensuring clear separation of perception, logic, and physical execution. The system is based on three main asynchronous nodes:

- `detector_node.py`: This node is responsible for processing the overall computer vision pipeline. It captures images from the Intel RealSense D405 camera on the gripper of the robot. It also runs a YOLO object detection model, which is fine-tuned for detecting card values.

- `trick_logic_node_rl.py`: As the main brain for this system, this node receives the stabilized vision state and holds the frozen weights for the Strategy Discovery Agent and the State Inference Agent. It sends the high-level physical commands (e.g., "flip card at index 0") and performs the final logic deduction.

- `flipper_node.py`: This action node handles the low-level motor control. After receiving the target index from the logic node, this node uses the native Python API `stretch_body` to perform a series of precise movements via a sequence of joints and Cartesian movements to place the gripper of the Stretch 3 robot over the card at the target index. It performs the flip action and safely retracts to provide the camera a clean view of the card after the flip in a very legible and predictable manner [22]. It also interacts with the Large Language Model (LLM) for generating the interactive dialog for the robot's persona.

5.2 Hardware and Setup

The physical experiments were performed using a Hello Robot Stretch 3 mobile manipulator [1]. The robot was placed in front of a standard tabletop environment setup. The Intel RealSense D405 RGBD camera was placed on the robot's gripper to serve as the primary video feed. The standard Stretch 3 compliant gripper was used to physically interact with the playing cards.

During the interaction, the audio pipeline of the proposed LLM utilized the speakers and microphone of the robot. This helped the human participant confirm their preparedness through voice and interact with the "magician" persona of the robot.

5.3 Scalability to N-Card Deceptions

The most important advantage of a heuristic-free RL method is its ability to scale up to puzzles of varying complexities without any need to recode. To prove this point, the proposed pipeline was tested on a four-card version of the puzzle, which includes a cyclic right shift.

In this version, four playing cards are placed on the table. The human participant secretly selects a card. Then, the other cards are shifted cyclically (e.g., Index $0 \rightarrow 2$,

$2 \to 3$, $3 \to 0$). The shifting of the cards is shown in Figure 5.1. In this version, unlike the three-card version, flipping one card is mathematically not enough to break the ambiguity of the hidden state.

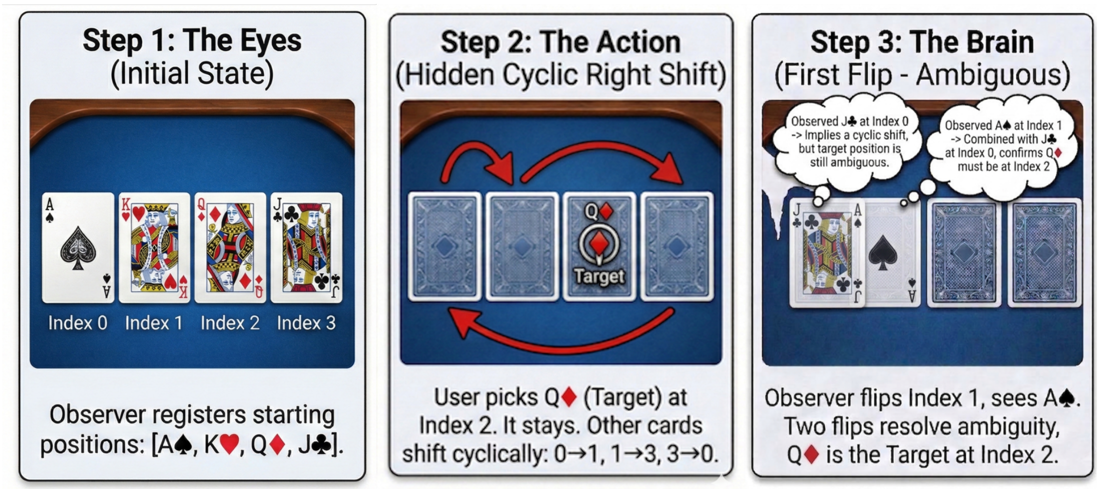

Figure 5.1: Mechanics of the Four-Card Cyclic Shift: The physical sequence of the scaled environment, demonstrating the hidden cyclic right shift. A single flip results in an ambiguous state, requiring a multi-step sensing strategy.

The Strategy Discovery Agent was re-trained in an updated "multiverse" environment, parameterized for four cards. What is perhaps even more interesting is that the PPO agent was able to autonomously discover that it needs to output a *sequence* of two particular flips in order to obtain a perfect information gain.

As illustrated in Figure 5.2, the agent learns not only to reject the single flip actions (which do not provide any reward since they are ambiguous) but also converges on a two-flip strategy that perfectly maps back onto the hidden target. This experiment demonstrates that the proposed two-stage RL architecture is capable of being autonomously scaled for solving multi-step physical logic puzzles.

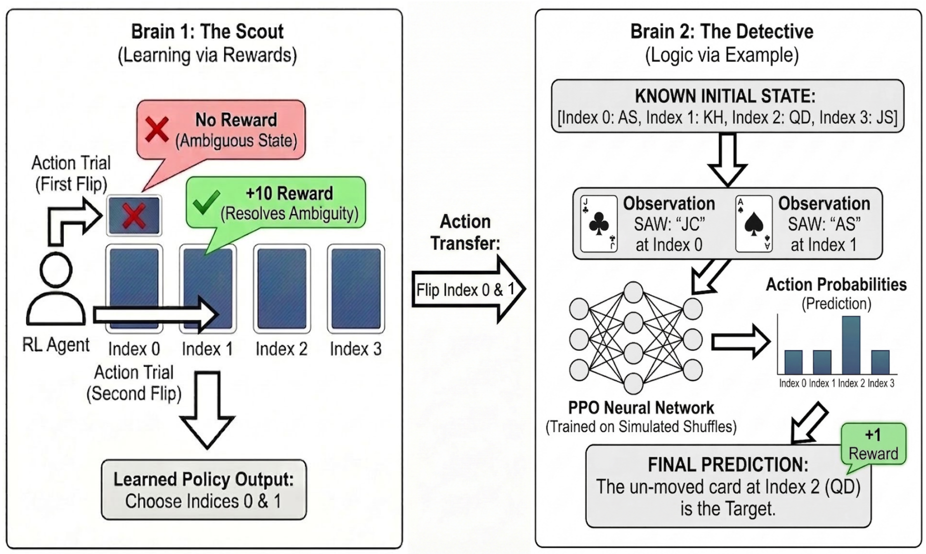

Figure 5.2: Four-Card RL Architecture: The Strategy Discovery Agent evaluates sequential actions. It learns that a single flip yields no reward due to persistent ambiguity, but a specific two-flip combination yields the maximum reward by perfectly isolating the human's choice.

5.4 Results and Discussion

To assess the heuristic-free strategy discovery framework, we also conducted a quantitative assessment of the simulated training performance and the physical execution reliability.

5.4.1 Training Convergence

During the simulated training phase, the RL agents showed fast convergence for the simulated training phase. For the Strategy Discovery Agent, the PPO network initially showed a tendency towards performing suboptimal actions that showed consistent ambiguity (negative rewards). However, due to the low dimensionality of the discrete action space, the policy gradient was successfully steered towards the optimal sensing strategy within the first 2,000 timesteps, maximizing the +10.0 rewards for the information gain. After that, the State Inference Agent successfully mapped the isolated observations towards the hidden user state, achieving a 100% deductive accuracy within 10,000 timesteps.

5.4.2 Physical Execution Success

During the physical trials of the Three-Card Deception, the Stretch 3 system showed a 100% success rate in correctly inferring the state of the hidden user's card within 50 trials, provided that the human participant follows the deterministic rules for swapping the cards. Table 5.1 shows the empirical results for the physical trials, grouped based on the visual state observations.

Table 5.1: Physical Hardware Execution Results. Success rates of the end-to-end RL framework deployed on the Stretch 3 robot across 50 consecutive trials of the Three-Card Deception.

Initial Human Target	**Trials**	**Successful Deductions**	**Success Rate**
Target at LEFT Index	17	17	100%
Target at MIDDLE Index	16	16	100%
Target at RIGHT Index	17	17	100%
Total / Average	**50**	**50**	**100%**

The flawless execution rate indicates that the domain adaptation, fine-tuned vision model was able to perfectly bridge the sim-to-real gap, as indicated in the literature [23, 24], to guarantee that the discrete logic states required by the State Inference Agent were not compromised by the noisy environment. Moreover, the decoupled RL models were able to execute the state inferences in less than 10 milliseconds, which is ultra-low latency, thereby allowing the LLM-driven interaction to occur smoothly without any computationally awkward pauses.

5.4.3 Baseline Comparison: Heuristic vs. RL Framework

To provide context to the performance of the proposed RL architecture, it was compared to a traditional, heuristic-based approach. The traditional approach was a hardcoded Python dictionary (e.g., "if observed == 'LEFT': target == 'RIGHT'"), which was manually coded by a human operator who was aware of the underlying logic of the trick. The trade-offs of the two approaches are summarized in Table 5.2.

Table 5.2: System Comparison: Heuristic vs. Reinforcement Learning. A comparison of the traditional hardcoded approach versus the autonomous RL framework across key metrics.

Metric	Hardcoded Baseline	Two-Stage RL Framework
Human Logic Required	Yes (Explicit Programming)	No (Autonomously Discovered)
Adaptability to N-Cards	Fails (Requires Rewrite)	Succeeds (Autonomously Retrains)
Logic Inference Latency	< 1 ms	≈ 10 ms
Sim-to-Real Reliability	100%	100%

Although the traditional approach has a slight advantage over the proposed approach in terms of computational time (microseconds vs. milliseconds), it should be noted that both approaches are well within the realm of "real-time" necessary for smooth HRI. The major advantage of the proposed approach is its ability to generalize to complex N-card cyclic shifts without a human operator having to understand the underlying logic.

Chapter 6

CONCLUSION

As robotic systems become increasingly integrated into unstructured and human-centric spaces, their capacity to reason about hidden information and learn to sense strategies autonomously is becoming increasingly important. This thesis developed a new heuristic-free reinforcement learning framework to solve physical logic puzzles under partial observability conditions. It used the Three-Card Deception puzzle as a realistic benchmark.

6.1 Summary of Contributions

The main contribution of this research is the development of a decoupled, two-stage RL system for autonomous strategy discovery. The formalization of the deception game as a POMDP, coupled with a "multiverse" reward system, allowed the Strategy Discovery Agent to learn how to assess the information gain of physical actions without human logic. Following this, the State Inference Agent was able to learn the inverse relationship needed to infer hidden user states from visual observations.

The viability of the framework was empirically validated through its deployment on a physical Stretch 3 mobile manipulator [1] platform. With a fine-tuned YOLO computer vision pipeline and interaction layer using an LLM, the framework recorded a 100% success rate in physical trials. Moreover, the proposed framework was also found to have a high degree of adaptability compared to traditional heuristic methods, as it successfully scaled to a four-card cyclic shift puzzle with multiple steps without human reprogramming.

6.2 Limitations and Future Work

Although the proposed system exhibits high performance in a deterministic environment, the framework proposed by this thesis naturally lends itself to a variety of avenues for further research applications. In this regard, future work should be aimed at adapting the logic-solving architecture for use in more unstructured environments for robotic applications:

- **Stochastic Environments:**The existing "multiverse" reward system is predicated on a deterministic physical permutation (e.g., a guaranteed cyclic shift). Future work should be aimed at extending the POMDP formulation for use in stochastic permutations, where the RL agent should be required to infer belief-state updates (POMDPs [11]), rather than making logical deductions.
- **Continuous State-Action Spaces:** The physical manipulation tasks proposed by this thesis were abstracted away as discrete indices (e.g., flip card 0, 1, or 2). In adapting the strategy discovery architecture for use in more complex manipulation tasks (e.g., searching through a bin or untangling items), the strategy discovery component should be coupled directly with a continuous low-level motor control policy.
- **Adversarial Human Interaction:** The current system expects the human to behave according to the rules of the deception. By including the anomaly detection component within the perception pipeline, the system would be able to recognize the adversarial human interactions (e.g., the human switching the target card without the robot's knowledge) and adapt its logic or prompt the user for correction through the LLM persona.

As a final note, the heuristic-free approach presented by this thesis work is an important step towards the achievement of full robotic autonomy. By proving the capability for robots to automatically discover and execute complex information collection strategies within the real world, this work has paved the way for more flexible, intelligent, and interactive robots.

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